RUTGERS

Phonetics II

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2026/11/11 (updated: 2026-11-07)

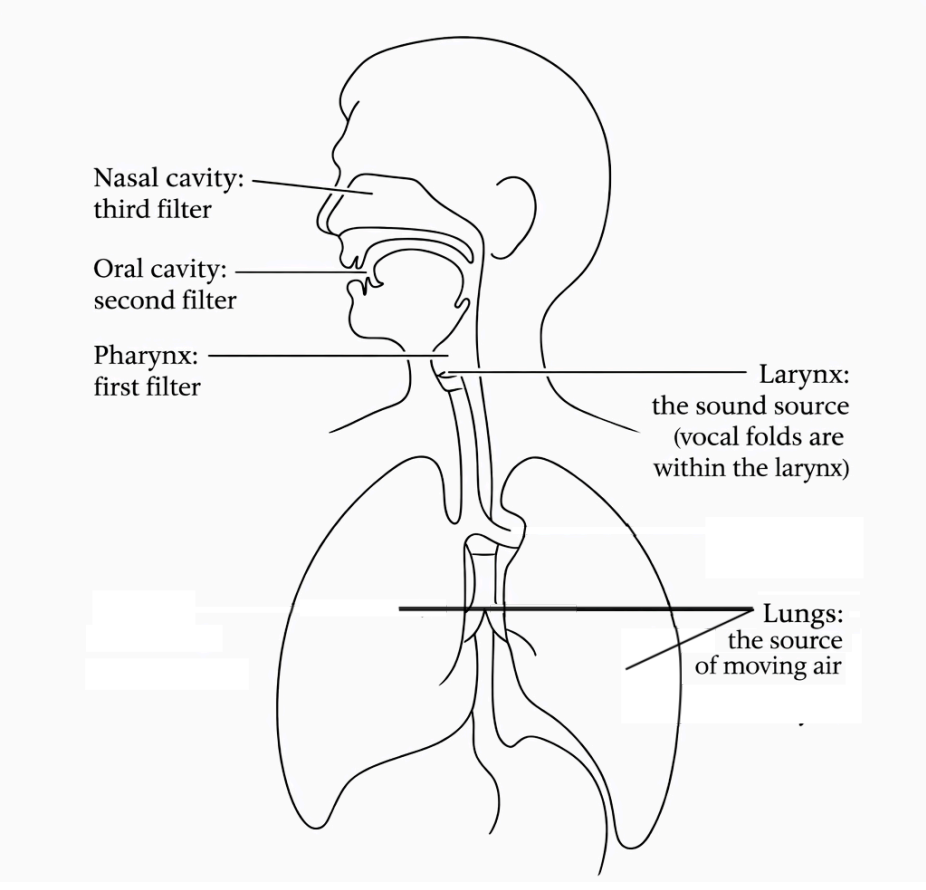
Please pick up your **name card** from the front

office hours: Mondays, 2:00-3:00 pm

Basement, Room 003, **Linguistics Building**

Previously on 201: Phonetics vs Phonology

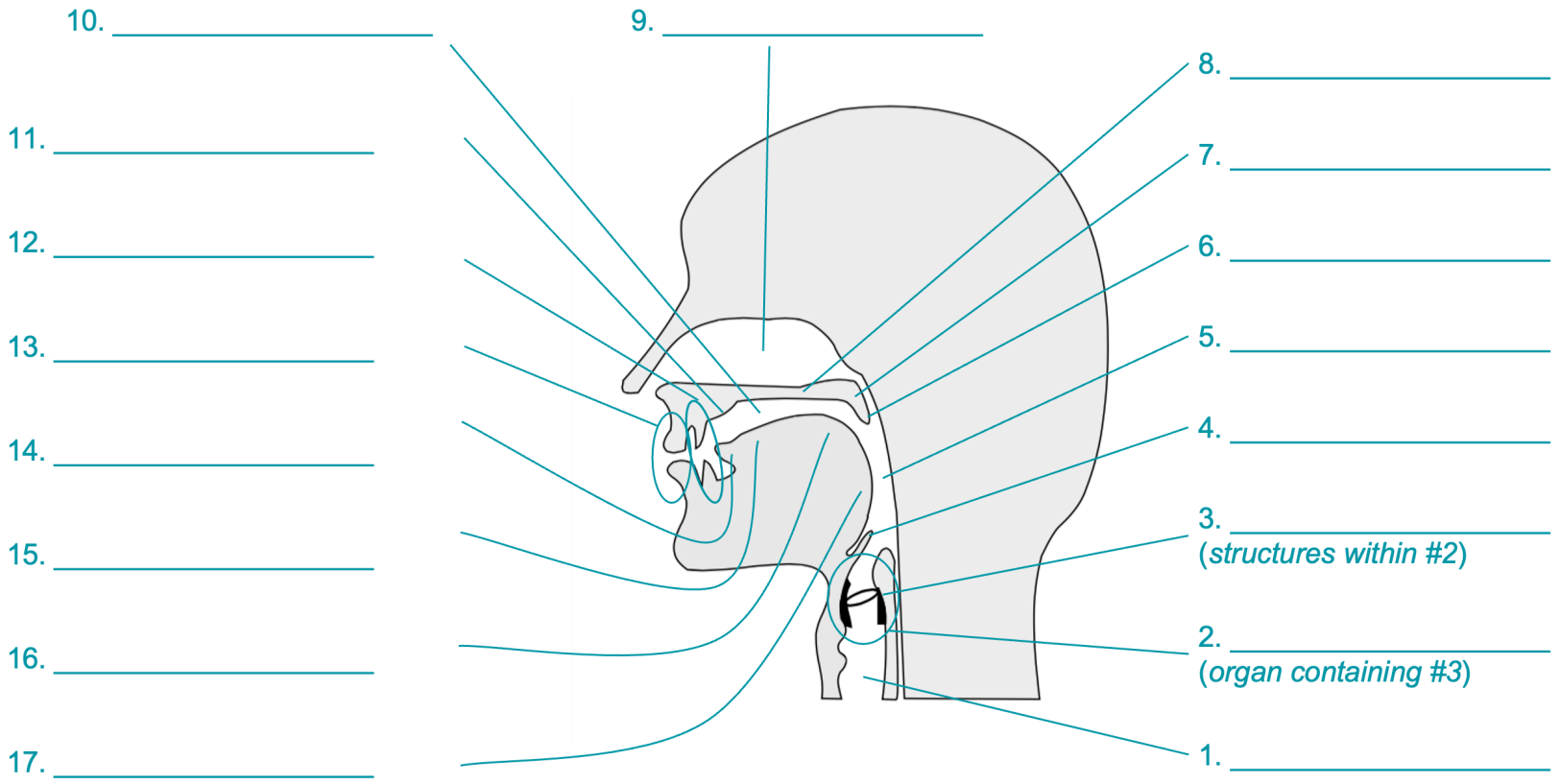
- **Phonetics** is the study of **how speech sounds are produced** by articulators in the vocal tract and **perceived** by are auditory apparatus
    - **articulatory phonetics**: how different sounds are produced in vocal tract
    - **acoustic phonetics**: the physical properties of speech sounds
    - **perceptual phonetics**: how sounds are perceived by listeners
- **Phonology** is the study of the **pattern** and **function** of sounds (how your brain categorises sounds associated with speech)



Vocal Tract Anatomy

alveolar ridge	nasal cavity	teeth	
epiglottis	oral cavity	tongue back	trachea
glottis & vocal folds	(hard) palate	tongue blade	uvula
larynx	pharynx	tongue root	velum (soft palate)
lips			

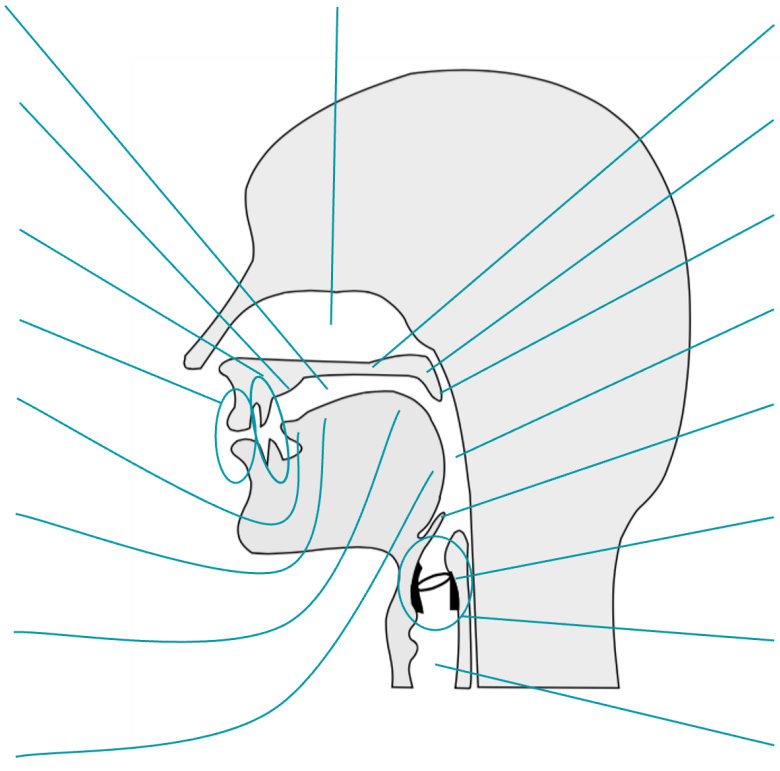
Fill in the blanks with the appropriate **anatomical term** for each **articulator** from the box. #5, 9, 10 refer to **cavities** (open spaces), not structures. (You may consult the reading, page 22)



Vocal Tract Anatomy

alveolar ridge	nasal cavity	teeth	
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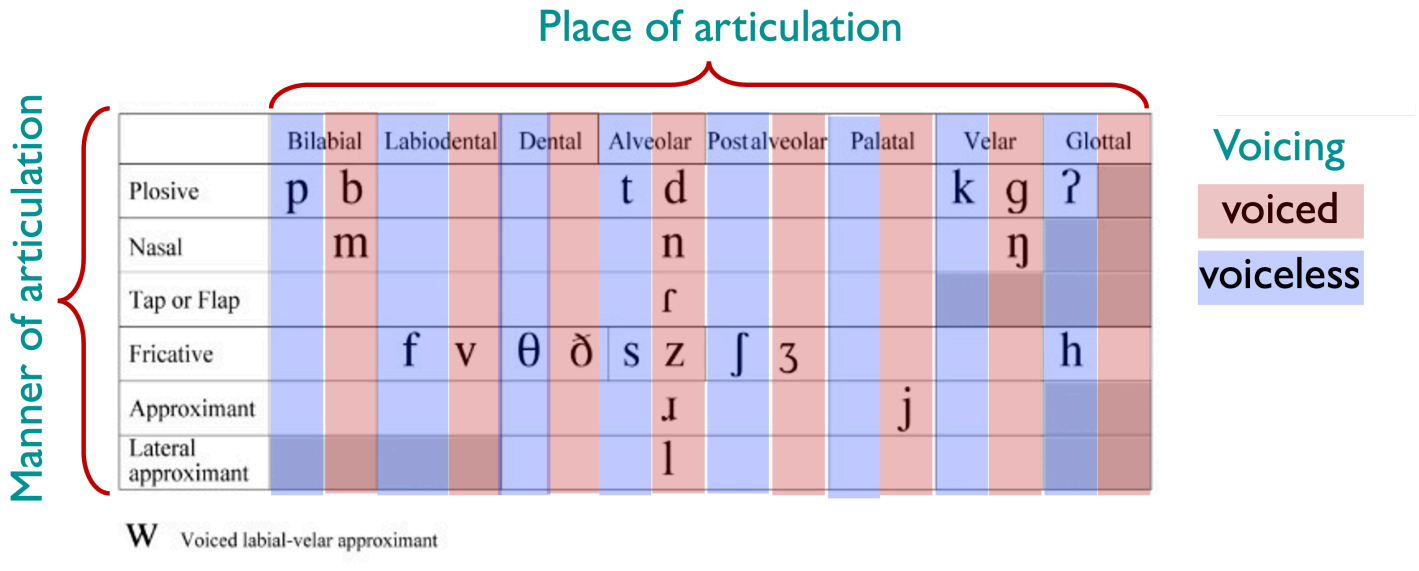


10. oral cavity
9. nasal cavity
11. alveolar ridge
12. teeth
13. lips
14. tongue tip
15. tongue blade
16. tongue back
17. tongue root
8. (hard) palate
7. velum (soft palate)
6. uvula
5. pharynx
4. epiglottis
3. glottis & vocal folds (*structures within #2*)
2. larynx (*organ containing #3*)
1. trachea

Previously on 201: Consonants

- all consonants are produced by restricting or obstructing the airflow in some way – whether partially or totally
- but the nature of the obstruction **varies** from consonant to consonant

- consonants can be characterized with three properties – which all serve to describe that obstruction
  1. **voicing**: presence or absence of vocal fold vibration
  2. **place of articulation**: where the obstruction of airflow takes place
  3. **manner of articulation**: how the airflow is modulated



Previously on 201: Feel Consonants

stops/plosives - **fricatives**: try to pronounce a **long** sequence of a consonant

- if it can be pronounced with non-stop, the sound is a **fricative**
- if it cannot be pronounced with non-stop, the sound is a **stop**



oral - **nasal**: hold **nose** and produce the sound

- if it becomes hard to produce, the sound is a **nasal**
- if it still sounds similar, the sound is a **oral**



try these tests on the first sound in < fan >, < pan > and < man >

Previously on 201: Voicing

(whether vocal cords vibrate or not)



Categories to Differentiate Consonants

CONSONANTS (PULMONIC) © 2020 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k g	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

columns - PLACE - where the obstruction of airflow in the vocal tract is

row - MANNER - what type of obstruction of airflow occurs

in a **single cell** - VOICING - whether the vocal cords vibrate or not (the right is *voiced*)

IPA clickable chart

Exercise I: Identifying Consonants

give a full description for **voicing**, **place**, and **manner** of **each consonant** in the following words and give the corresponding **IPA** symbol

move

cloth

gosh

hide

stove

bank

CONSONANTS (PULMONIC) © 2020 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
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Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

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Exercise I: Identifying Consonants

give a full description for **voicing**, **place**, and **manner** of **each consonant** in the following words and give the corresponding **IPA** symbol

move
[m]: voiced bilabial nasal
[v]: voiced labio-dental fricative

cloth
[k]: voiceless velar stop
[l]: voiced alveolar lateral approximant
[θ]: voiceless dental fricative

gosh
[g]: voiced velar stop
[ʃ]: voiceless post-alveolar fricative

hide
[h]: voiceless glottal fricative
[d]: voiced bilabial stop

stove
[s]: voiceless alveolar fricative
[t]: voiceless alveolar stop
[v]: voiced labio-dental fricative

bank
[b]: voiced bilabial stop
[ŋ]: voiced velar nasal
[k]: voiceless velar stop

Practice: Identifying Consonants

give a full description for **voicing**, **place**, and **manner** of **each consonant** in the following words and give the corresponding **IPA** symbol

CONSONANTS (PULMONIC) © 2020 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k g	q ɢ		ʔ
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Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

Vowels and Consonants

- the way we articulate vowels and consonants differs in two important ways

- **airflow** – when we speak, we exhale, forcing air to flow outward through our **vocal tract**
  - with **consonants**, the airflow is blocked completely or partially
  - with **vowels**, there is no obstruction

- **voicing** – when we cause our vocal cords to vibrate during speech
  - **vowels** are usually voiced
  - **consonants** may be voiced or voiceless

	Vowels	Consonants
Obstruction of airflow	No obstruction in vocal tract; air flows freely	Total or partial obstruction in vocal tract, blocking or restricting airflow
Voicing	Usually voiced	May or may not be voiced

Voicing and the Larynx

- **larynx** is the primary source of speech sounds and it contains
  - **vocal folds** (vocal cords): two sheets of muscle that can be brought together or pulled apart
  - **glottis**: opening between the vocal folds through which air passes
  - other structures of muscle, cartilage, and ligament that **support** and **move** the vocal folds

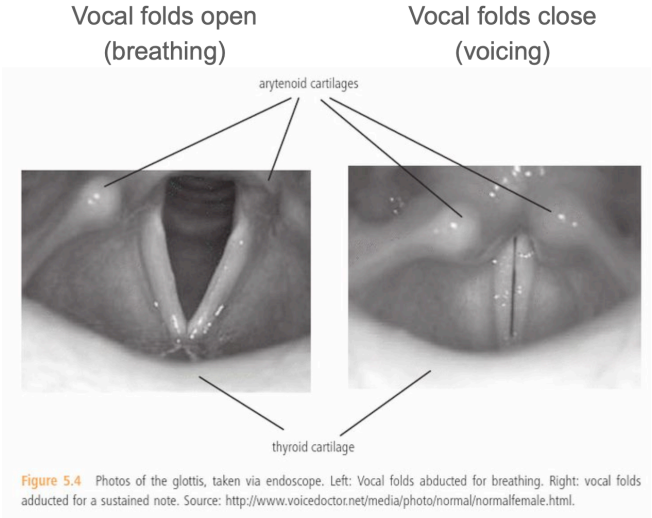


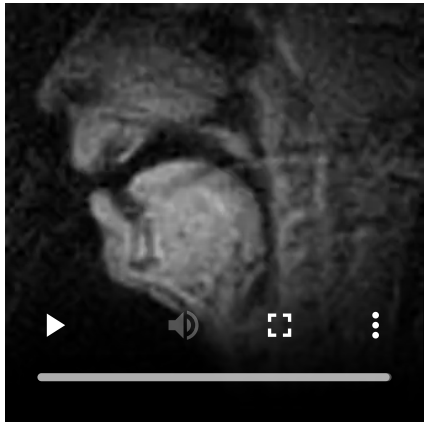
Figure 5.4 Photos of the glottis, taken via endoscope. Left: Vocal folds abducted for breathing. Right: vocal folds adducted for a sustained note. Source: http://www.voicedoctor.net/media/photo/normal/normalfemale.html.

video: the larynx of an opera singer during breathing and phonation

- if vocal folds are **abducted close together** and the air comes out, they'll **vibrate** due to **Bernoulli's effect**
  - the **vibration** leads to **voicing** and is used in vowels [a, e, i, o, u] and certain consonants [b, d, g]
- if vocal folds are **moved apart**, they'll stop vibrating, leading to **voicelessness** [p, t, k]

Feel the Vowels

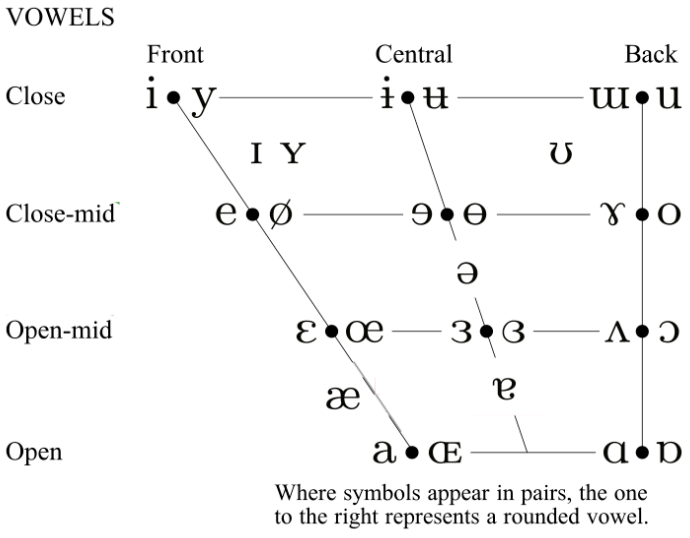
- three different vowels were produced, and the processes have been captured with **real-time MRI**
- the three vowels are **[i]**, **[u]**, **[a]**, as in < st**ee**p >, < st**oo**p >, < st**o**p >
- which one is which?







Feel the Vowels



- three different vowels were produced, and the processes have been captured with **real-time MRI**
- the three vowels are **[i]**, **[u]**, **[a]**, as in < st**ee**p >, < st**oo**p >, < st**o**p >
- it's no coincidence that our **oral cavity** and **the vowel chart** resemble
- the relevant articulator for vowels is the **tongue**, which **shapes** the air flow



[a] in < st**o**p >



[i] in < st**ee**p >



[u] in < st**oo**p >

Cardinal Vowels

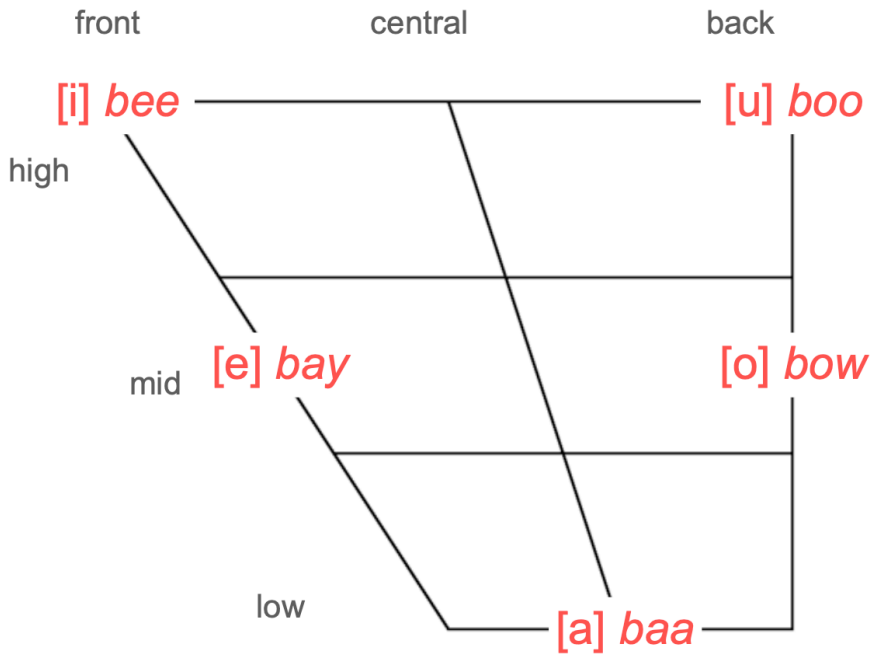
- **vowels** are articulated with
  1. constant airflow (NO **obstruction**)
  2. voicing (**vibration** of vocal folds)

- it's useful to establish a few simple vowels – **cardinal vowels** – and build up from there

- 5 cardinal vowels of Spanish is in the chart on the right, and many other languages have a similar 5-vowel system

- English key words are attached with the vowel to show roughly how these vowels sound

- **note** that the International Phonetic Alphabet's symbols for vowels don't always match our English names for them

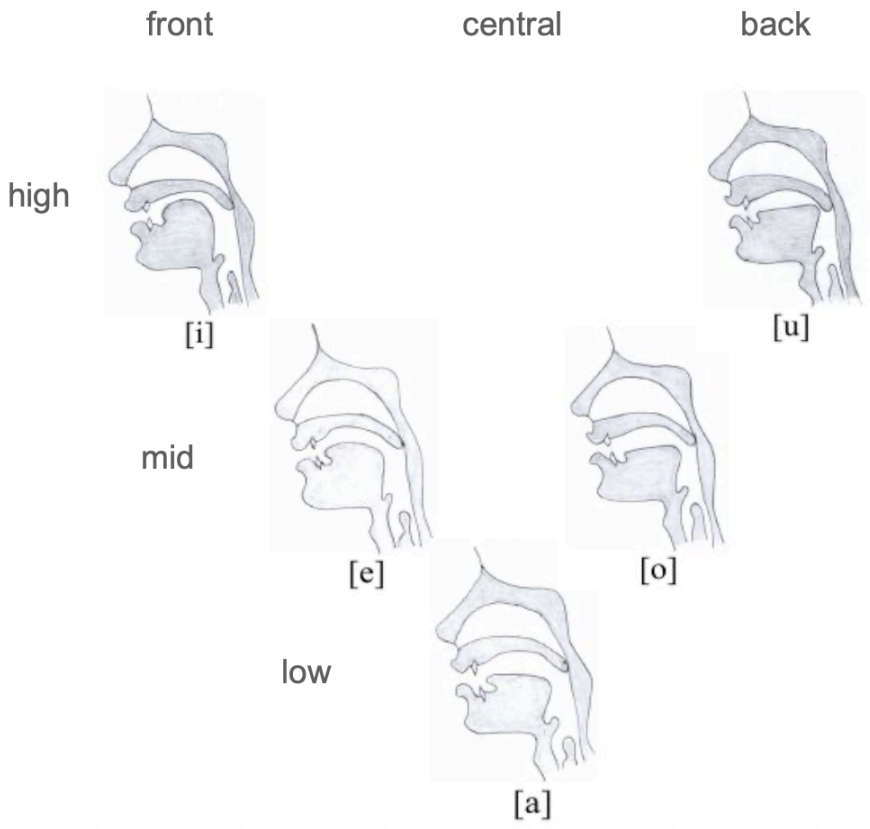


Spanish Vowels

Vowel Articulation

different vowel qualities can be primarily be distinguished based on **four** features

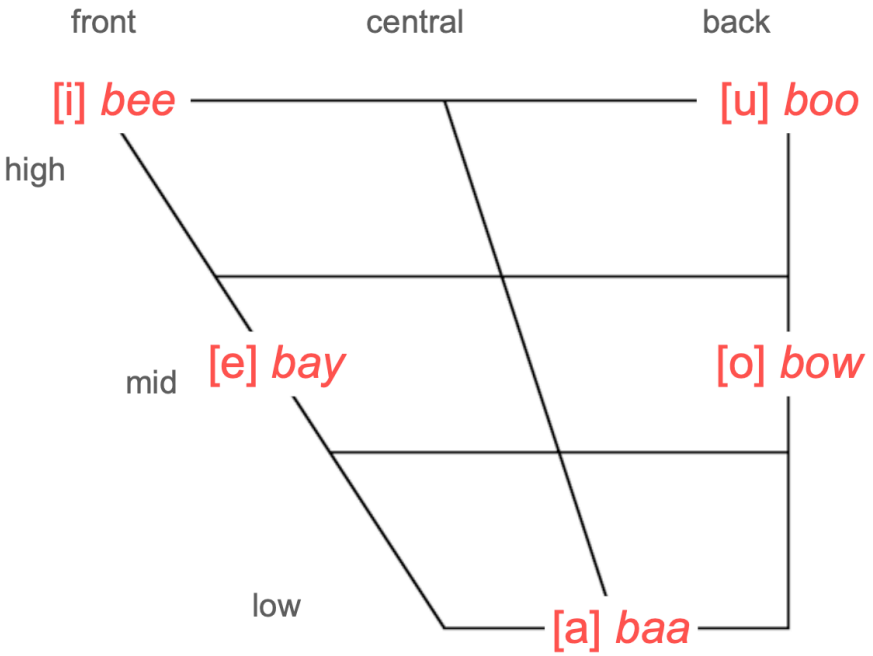
1. tongue **height**
   - high, mid, low
2. tongue **backness**
   - front, central, back
3. lip **roundness**
   - rounded, unrounded
4. **tenseness**
   - tense and lax (**lax**: relaxed tongue)
   - English-specific
   - ATR (advanced tongue root)



Vowels in a Real-time MRI

- the link below is to a video that is produced by an **MRI** (magnetic resonance imaging) scanner
- watch how the tongue and lips move as these people sing [a, e, i, o, u]

vowel MRI

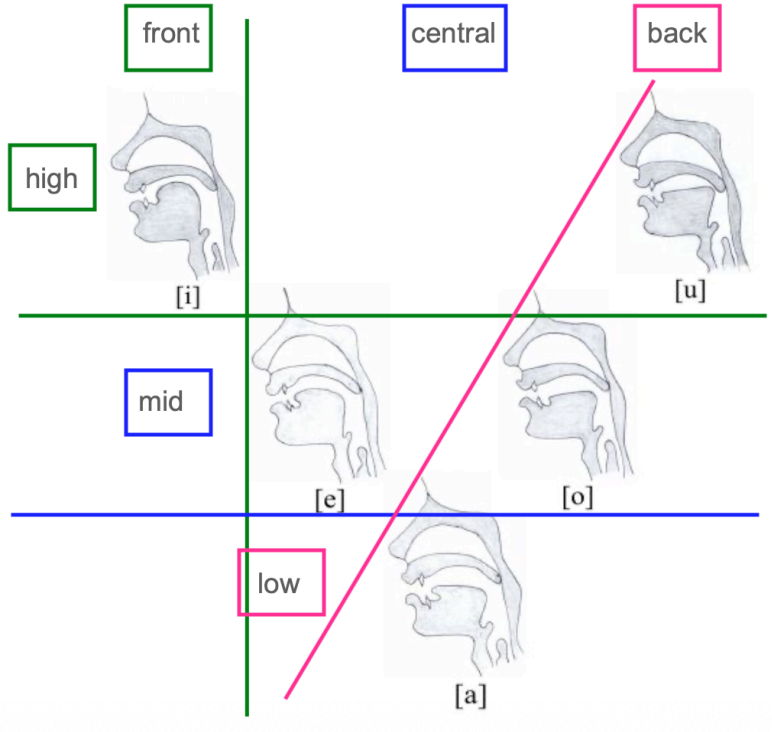


Spanish Vowels

Vowel Features

- in different vowel qualities, we can distinguish various levels of **tongue height**
  - high [+ high, - low]
  - mid [- high, -low]
  - low [- high, +low]
  - why don't we use **high** or **low** only

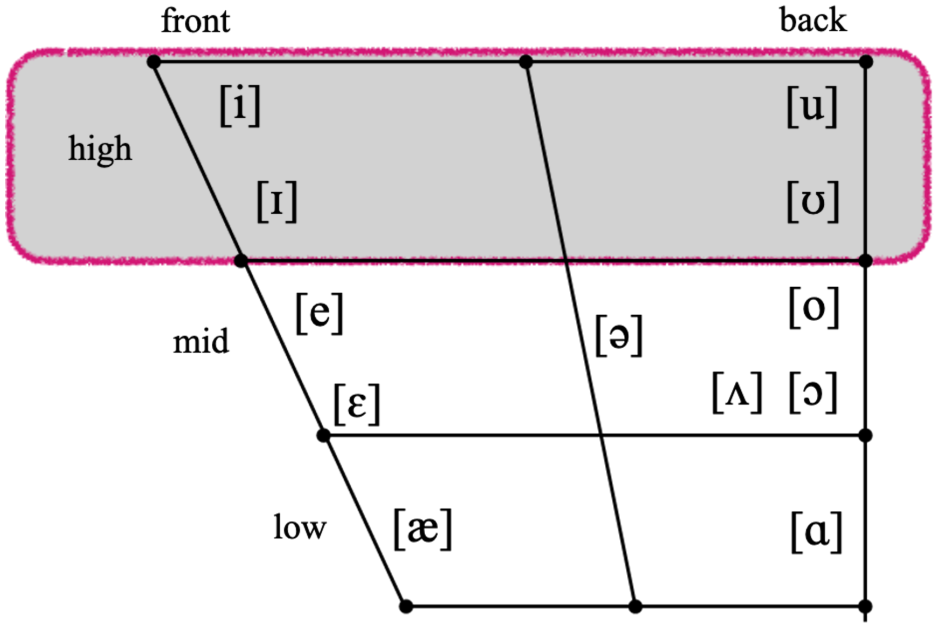
- and we can distinguish various levels of **tongue backness**
  - front [+ front, - back]
  - central [- front, - back]
  - back [- front, + back]



- for English, we actually do not have **central** vowels, so using only **back** is enough

- arranging them by tongue position, they form a triangle, so vowels are often arranged in a **vowel triangle/trapezoid**

Height of the Tongue: American English

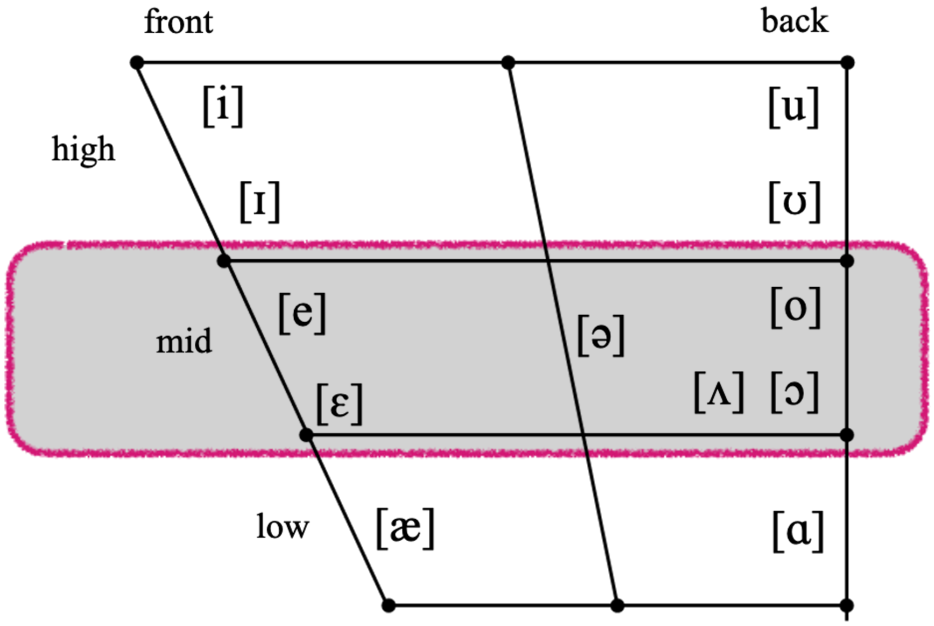


high vowels

[+high, -low]

< b**ee**t >	[i]
< b**i**t >	[ɪ]
< b**oo**t >	[u]
< b**oo**k >	[ʊ]

Height of the Tongue: American English

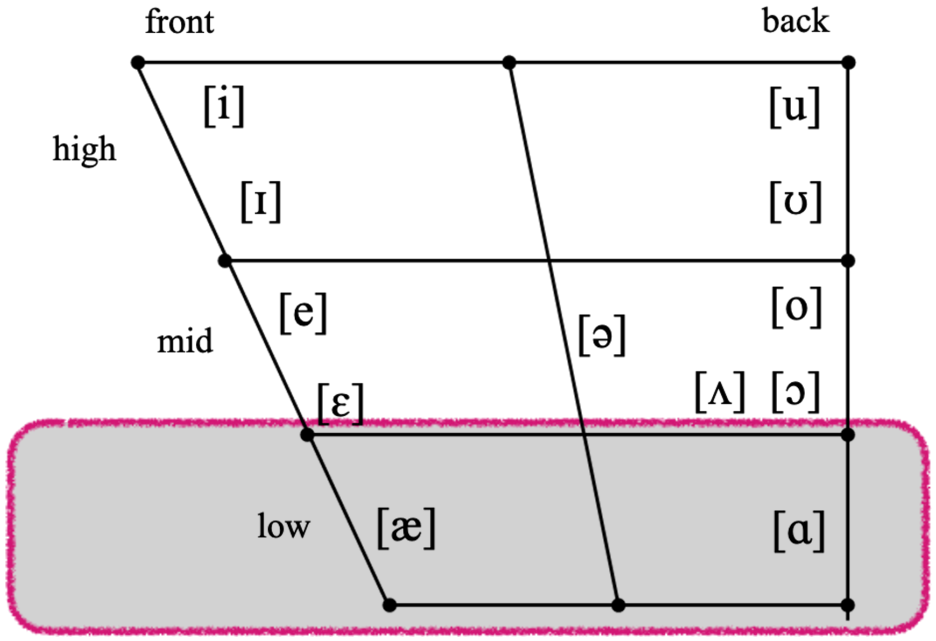


mid vowels

[-high, -low]

< b**e**t >	[ɛ]
< b**ai**t >	[eɪ]
< **a**bout >	[ə]
< b**u**t >	[ʌ]
< b**oa**t >	[oʊ]
< b**ough**t >	[ɔ]

Height of the Tongue: American English



low vowels

[-high, +low]

< b**a**t > [æ]

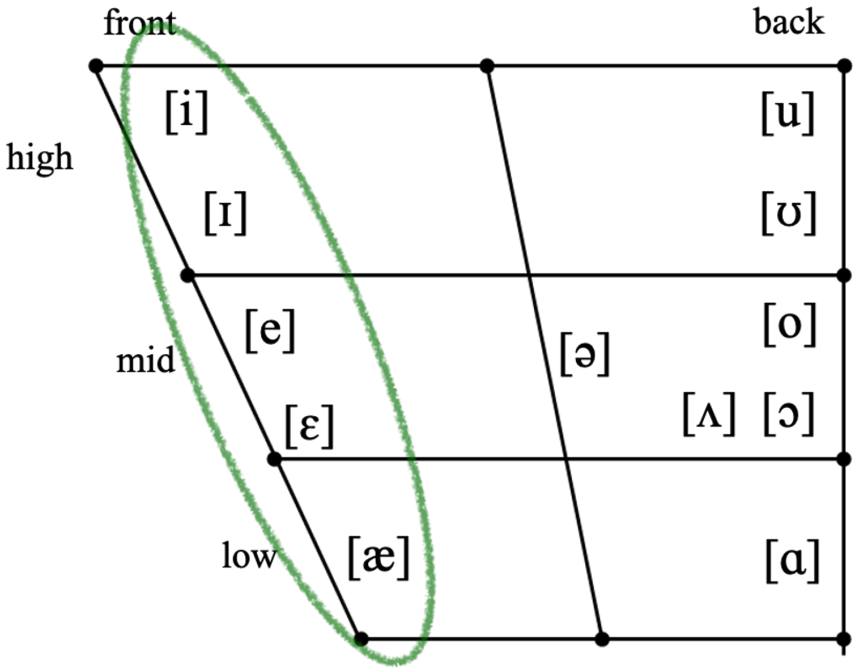
< b**o**t > [ɑ]

Backness of the Tongue: American English

front vowels

[+front, -back]

< b**ea**d >	[i]
< b**i**d >	[ɪ]
< b**a**de >	[eɪ]
< b**e**d >	[ɛ]
< b**a**d >	[æ]

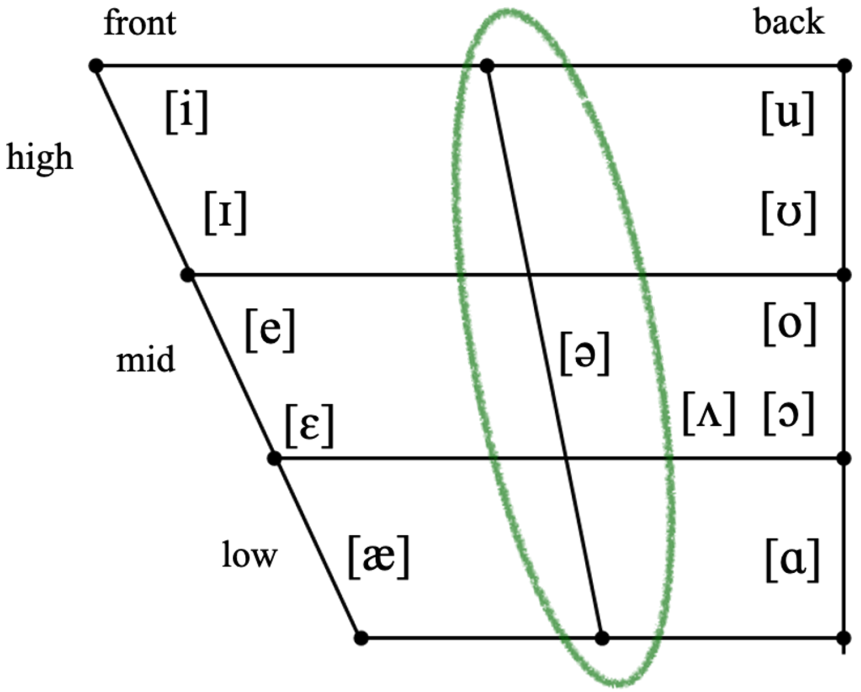


Backness of the Tongue: American English

central vowels

[-front, -back]

< about > [ə]

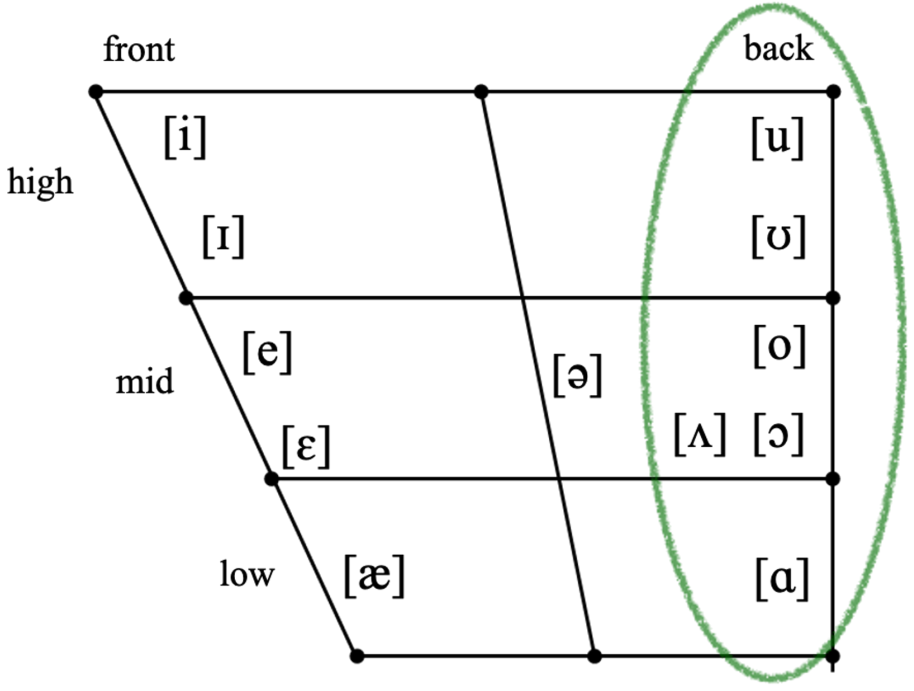


Backness of the Tongue: American English

back vowels

[-front, +back]

< b**oo**t >	[u]
< p**u**t >	[ʊ]
< b**oa**t >	[oʊ]
< b**ough**t >	[ɔ]
< b**u**t >	[ʌ]
< b**o**t >	[ɑ]



Lip Rounding

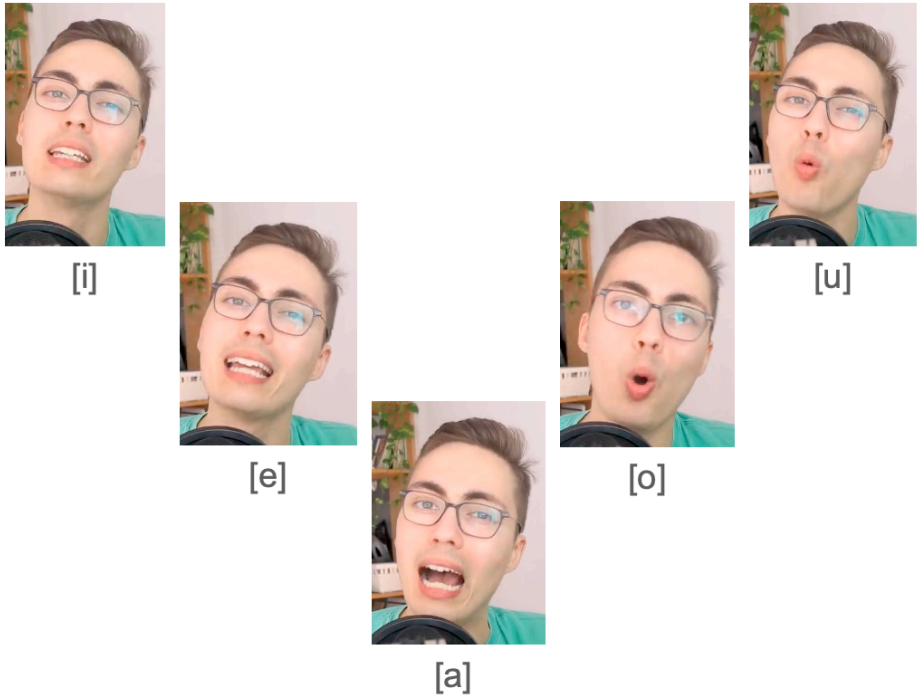


image source: *Podcast Italiano* - Le vocali dell'italiano standard

- the third vowel feature is **lip rounding**
  - rounded [+ round]
  - unrounded [- round] (spread)
- in the link on the right, there are our **five vowels** as pronounced by a speaker of Italian
  - how does his mouth shape change for each vowel?
  - which ones are rounded/unrounded?
- **note** that he's **exaggerating** a bit for the purposes of demonstration
  - since in the video he's trying to **teach** Italian pronunciation

Lip Rounding

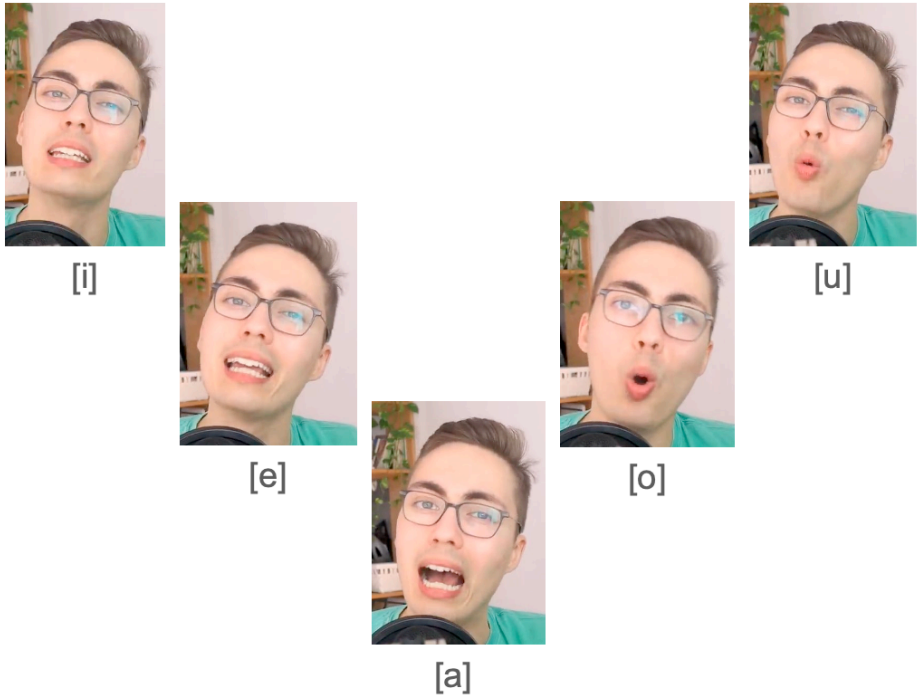
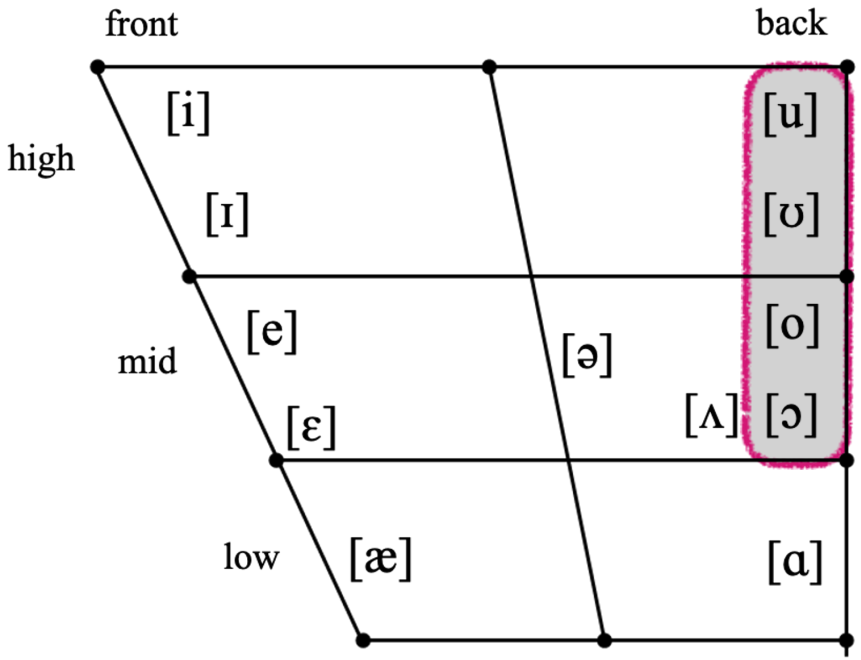


image source: *Podcast Italiano* - Le vocali dell'italiano standard

- [o, u] are **rounded**; [i, e, a] are unrounded
- which follows a **crosslinguistic tendency**
- **back** vowels tend to be **rounded**
  - **front** and low vowels tend to be **unrounded**
- front rounded vowels exist but are **less common**
  - **German**: *schön* [ʃøːn] 'beautiful'
  - **Kazakh**: *ypy* [yru] 'to bark'
- **as are** back unrounded vowels
  - **Japanese**: *kuzu* [kɯzɯ] 'kudzu vine'

Roundness of the Lips: American English



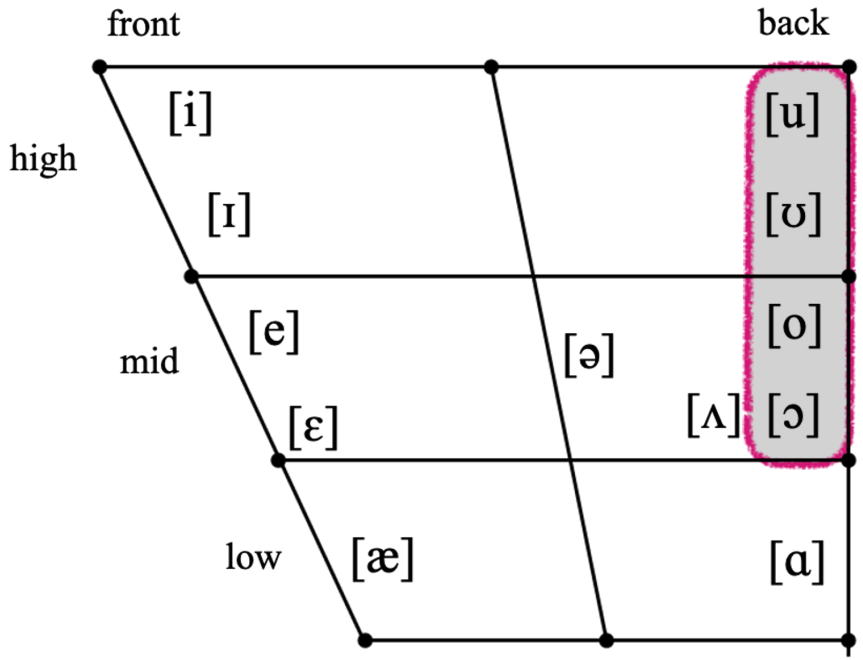
rounded vowels

[+round]

< p**u**t >	[ʊ]
< b**oo**t >	[u]
< b**oa**t >	[oʊ]
< b**ough**t >	[ɔ]

- the rest of the vowels are **unrounded** ([-round])

Roundness of the Lips: American English



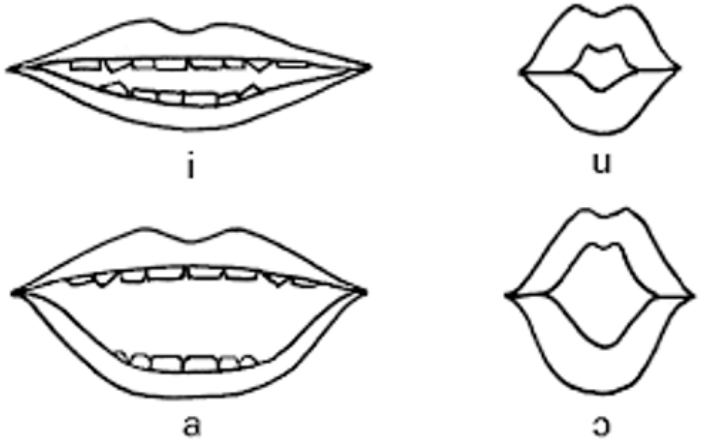
match the words with the images for vowels

(a) < **mean** >

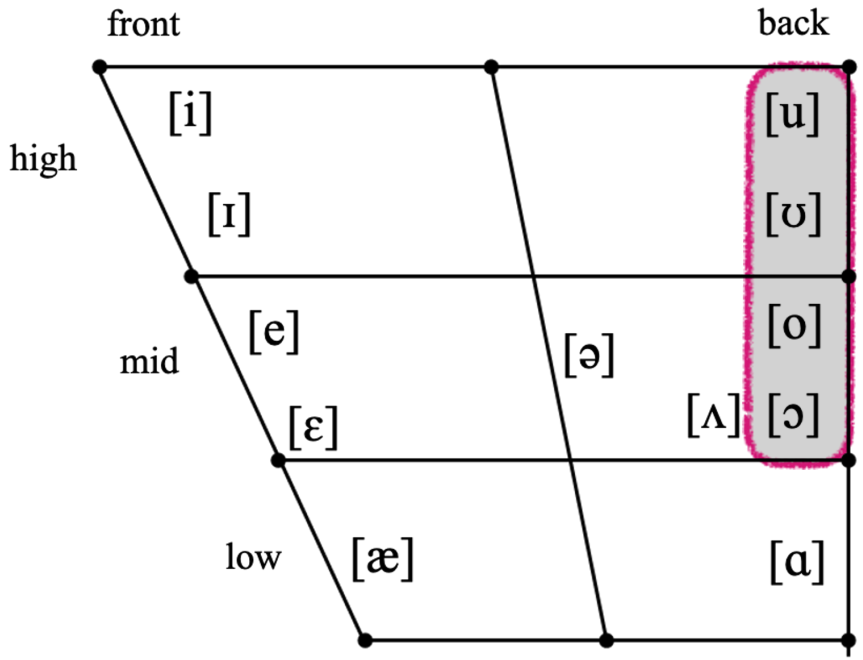
(b) < **moon** >

(c) < **mine** >

(d) < **moth** >



Roundness of the Lips: American English



match the words with the images for vowels

(a) < **mean** >

(b) < **moon** >

(c) < **mine** >

(d) < **moth** >

[i]



[u]





[ɑj]

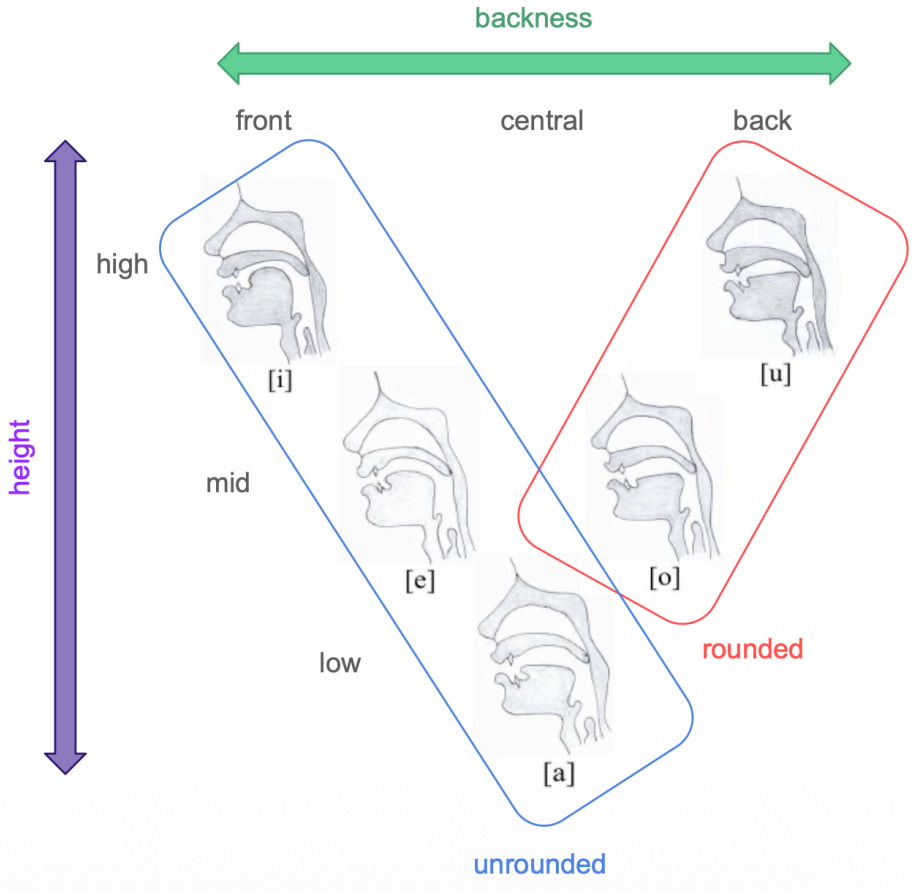


[ɔ]

Vowel Feature

- so these are the three **articulatory features** we use to distinguish different vowel qualities:
    - **tongue height**: how high is the tongue in the mouth
    - **tongue backness**: is tongue further forward or back in mouth
    - **lip roundedness**: are the lips rounded or unrounded

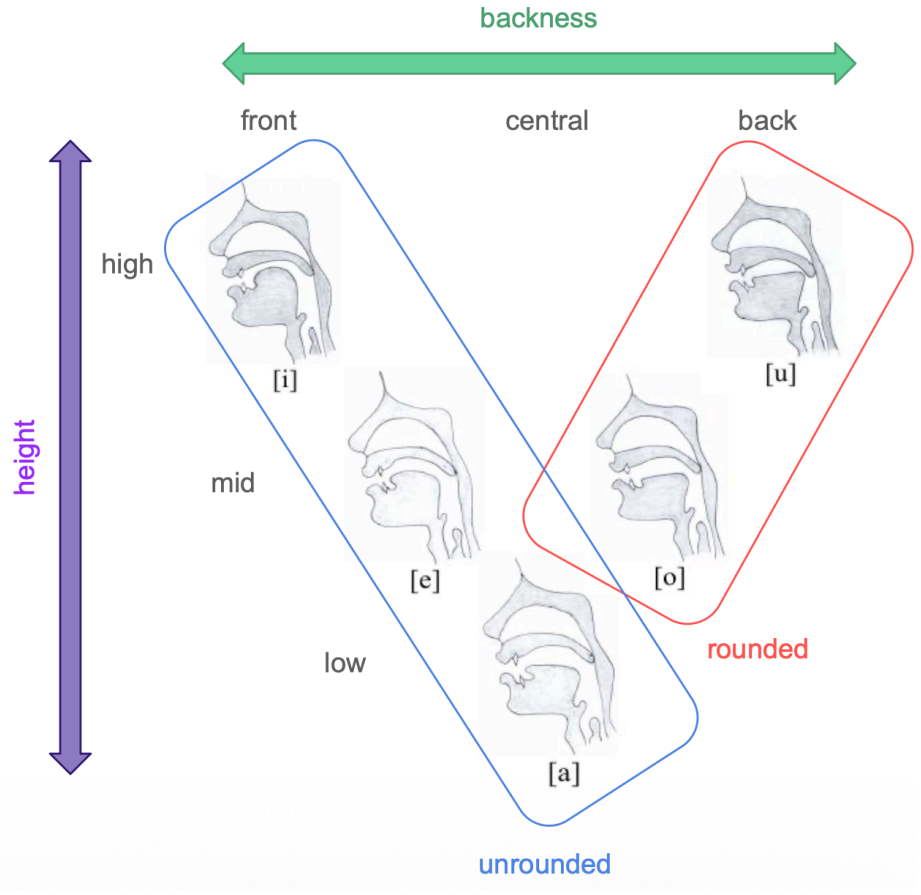
- when we describe vowels, we typically do so in terms of **height** - **backness** - **roundedness**
    - **[i]** =
    - **[e]** =
    - **[a]** =
    - **[o]** =
    - **[u]** =



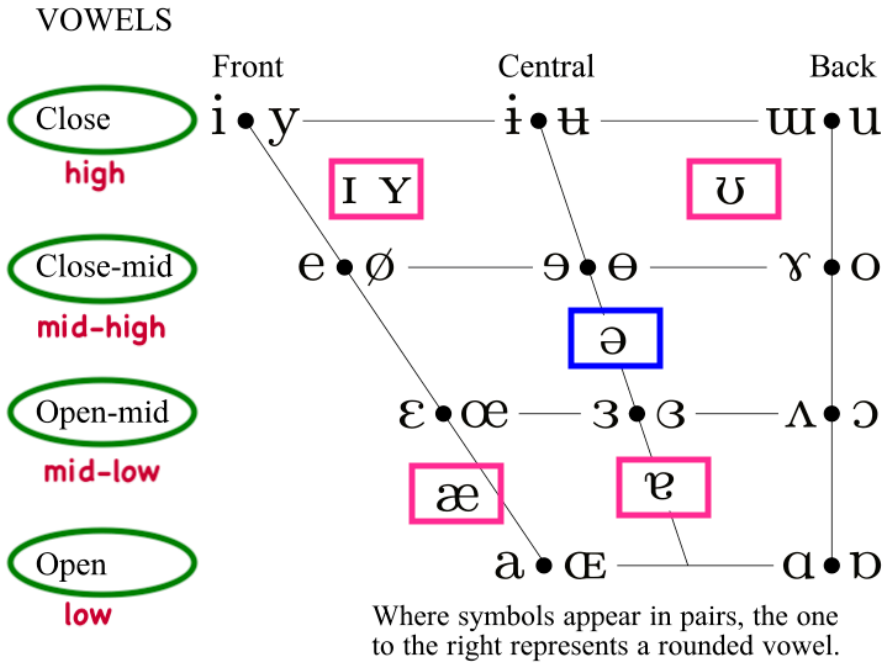
Vowel Feature

- so these are the three **articulatory features** we use to distinguish different vowel qualities:
  - **tongue height**: how high is the tongue in the mouth
  - **tongue backness**: is tongue further forward or back in mouth
  - **lip roundedness**: are the lips rounded or unrounded

- when we describe vowels, we typically do so in terms of **height** - **backness** - **roundedness**
  - **[i]** = high front unrounded vowel
  - **[e]** = mid front unrounded vowel
  - **[a]** = low central unrounded vowel
  - **[o]** = mid back rounded vowel
  - **[u]** = high back rounded vowel



Beyond the Cardinal Vowels

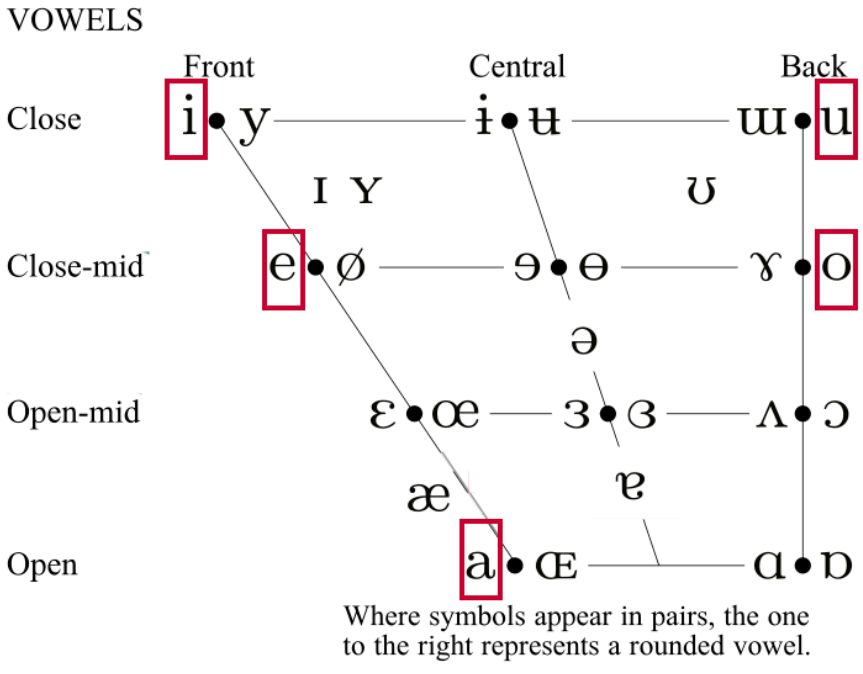


- IPA chart uses **28 cardinal vowels** to try to represent all of the world's languages – though **no language uses them all**

- **some notes**

  1. IPA chart uses terms **close/open** instead of **high/low**, but mostly we use **high/low** *[close = high; clode-mid = mid-high, etc]*
  2. IPA chart distinguishes **4 levels of height** instead of 3
  3. IPA chart includes some **vowels between these levels** [ɪ, ʏ, ʊ, æ, ɐ] and terms like near high, near front are used for them
  4. **[ə]** (schwa) is in the **center** of the chart and is the mid central vowel

Beyond the Cardinal Vowels

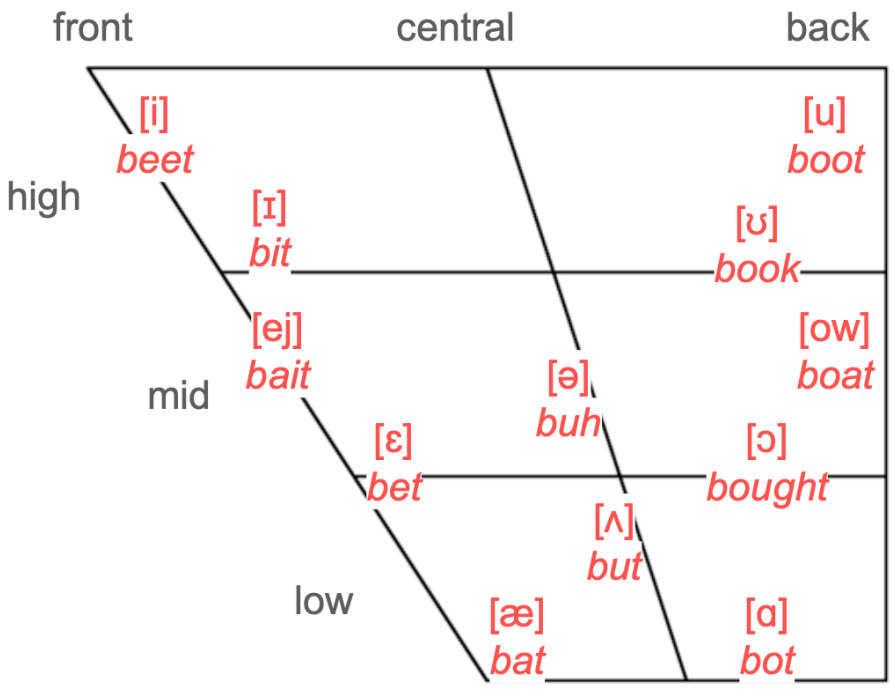


- how do we figure out what sound these symbols **represent**?
- **five cardinal vowels** as a starting point
- how is [y] pronounced?
  - start with [i]and make it rounded
- how is [ɛ] pronounced?
  - tongue about halfway between [e] and [a]
- how is [ɨ] pronounced?
  - tongue about halfway between [i] and [u]
- how is [ɵ] pronounced?
  - tongue about halfway between [e] and [o]

American English Vowels Summary

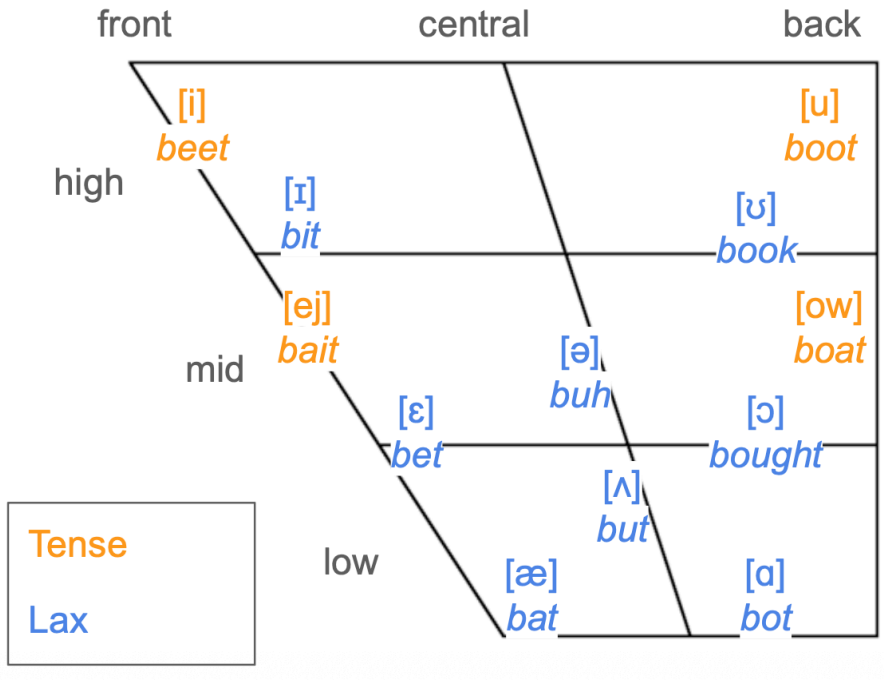
- American English has around **12-14** vowels
- exact number and pronunciation varies by region – English accents **tend to differ** in terms of vowels
- a few **notes** about American English vowels

1. there is a 5-way **height** contrast
2. Speakers tend to **diphthongize** some of these vowels, especially the vowels in *bait* and *boat*, but also to a certain degree in *beet* and *boot*
3. some American English speakers don't differentiate [ə] and [ʌ], and some don't differentiate [ɑ] and [ɔ]

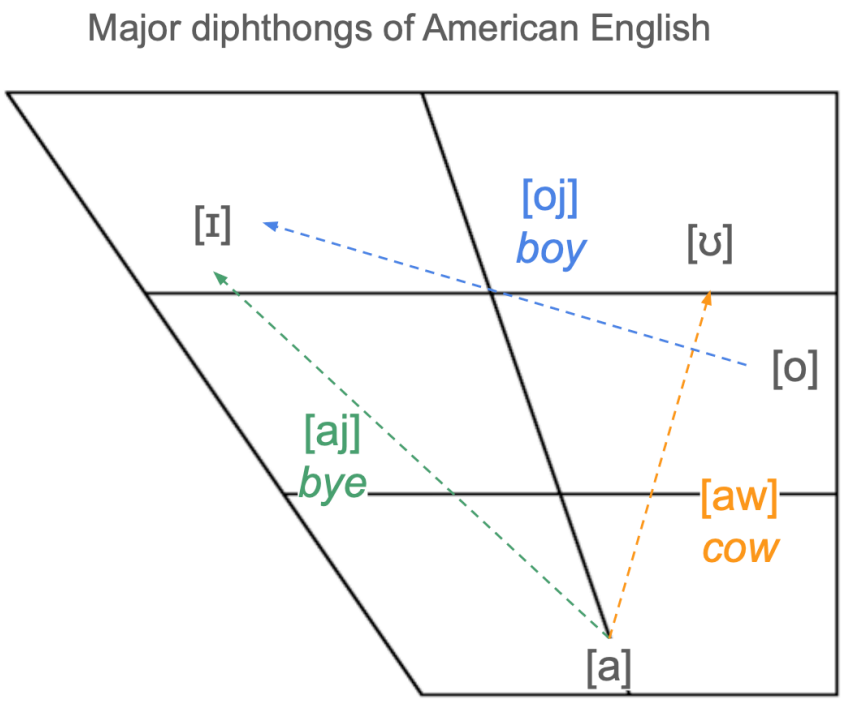


Tense and Lax Vowels

- 5 height differences is a bit unwieldy
- a common solution is to divide the vowels into **two groups**
- **tense** vowels
  - [i], [u], [eɪ̯] (or [ej]), [oʊ̯] (or [ow])
  - a bit higher and more peripheral, tend to be diphthongized in American English
- **lax** vowels
  - the rest
  - tongue a bit more relaxed
  - **cannot end a word** in American English



Diphthongs and Glides



- a diphthong is **two vowel sounds** that are combined into a single sound
  - **vowel** + **glide** or **glide** + **vowel**
- a **glide** is considered **semi-vowel**
  - it is a vowel-like sound produced by moving your tongue while saying a vowel
- by combining a glide with a vowel, we get a **diphthong**
  - the tongue is gliding around the mouth as you make a diphthong
- there also exist triphthongs
  - **glide** + **vowel** + **glide**
  - *yay*, *wow*, *way*

Other Vowel Features



nasalisation: articulated by lowering the velum/soft palate, so that air flows through both the **nasal** and **oral** cavity

- **English**
  æt '*at*' **vs.** æ̃nt '*ant*'

length: long vowels **contrast** with short vowels

- **Mongolian**
  [ʦɑˈsɯn] '*snow*' **vs.** [ʦɑɑˈsɯn] '*paper*'
- since adding or removing the **length feature** changes the meaning completely
- we say that length is **contrastive** in mongolian

pitch: tonal languages **contrast** pitch

- **Mandarin Chinese**

ma^{55}	ma^{35}	ma^{214}	ma^{51}	ma^{00}
'*mom*'	'*numb*'	'*horse*'	'*scold*'	'*Q-mark*'

Assignment VI is due this Friday (**Nov 13th**)

Assignment VII will be published today (**Nov 11**) and is due next Friday (**Nov 20th**) **continue reading**:
Phonetics in O'Grady et al.'s *Contemporary Linguistics: An Introduction*

start reading for phonology:
Phonology: The Function and Patterning of Sounds
in O'Grady et al.'s *Contemporary Linguistics: An Introduction*

This is Schwa.

Schwa is not stressed.

Schwa is cool.

Be like Schwa.





Slides created via the R package **xaringan**.